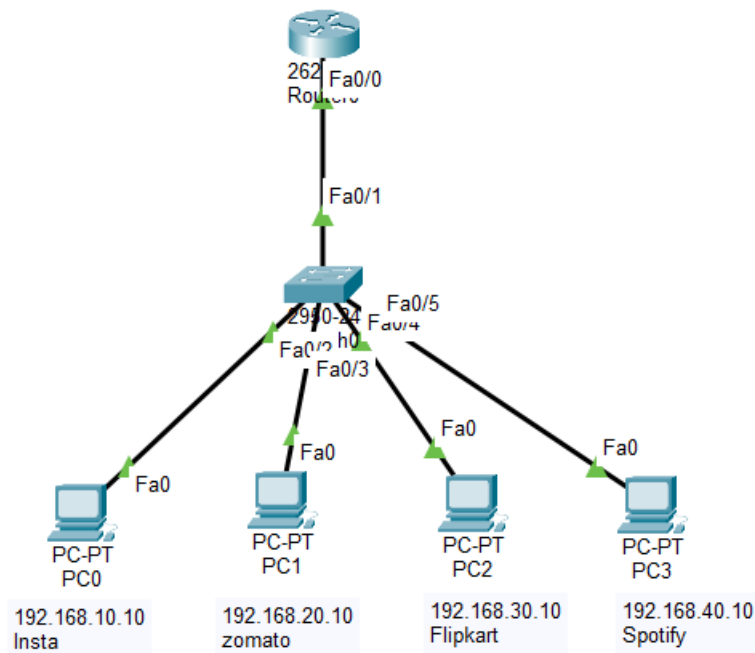


Inter-VLAN Routing Concepts???? Theory: Why is Inter-VLAN Routing Needed? Problem: VLANs cannot communicate with each other Solution: Use a router or Layer 3 switch Methods of Inter-VLAN Routing: Router-on-a-Stick (RoS) (Single router interface for multiple VLANs) Layer 3 Switch Routing???? Practical: Router-on-a-Stick Configuration???? Hands-on Lab Enable router sub-interfaces for VLANs

Task 1.

On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.



Task 2.

Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10 Insta	active	Fa0/2
20 Zomato	active	Fa0/3
30 Flipkart	active	Fa0/4
40 Spotify	active	Fa0/5
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

Task 3.

Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command.

Hint: Use the switchport mode trunk command on the relevant interface.

```
Switch>
Switch>en
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40

Switch#
```

Task 4.

Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=4ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```

- The ping from the Insta VLAN PC to the Zomato VLAN PC was successful because Router-on-a-Stick was configured correctly.
- The switch port connected to the router was configured as a trunk, allowing VLAN traffic to pass between the switch and router.
- The router sub interfaces Fa0/0.10 and Fa0/0.20 acted as the default gateways and routed traffic between VLAN 10 and VLAN 20.

Task 5.

Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=4ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```

Switch

vlan 40

name Spotify

exit

do wr

```
interface fa0/5  
switchport mode access  
switchport access vlan 40  
exit  
do wr
```

Router

```
interface fastEthernet0/0.40  
encapsulation dot1Q 40  
ip address 192.168.40.1 255.255.255.0  
exit  
do wr
```